

Predicting Customer Churn in Financial Services Using Machine Learning

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Abstract—Customer churn remains a major challenge for financial institutions because losing customers affects long-term profitability and increases acquisition costs. This research develops a machine learning framework for predicting customer churn in banking environments by combining Logistic Regression, Random Forest, XGBoost, and Neural Network models. To improve prediction quality, the framework incorporates SMOTE for class balancing, behavioural feature engineering, and SHAP-based explainability. Experiments conducted on the Bank Customer Churn Dataset demonstrate that the Hybrid Ensemble model achieves superior overall performance with an F1-score of 0.5997 and an ROC-AUC of 0.8471. Feature importance analysis reveals that Age, Balance, and Active Membership status are among the most influential factors affecting customer churn. The proposed framework provides both predictive capability and business interpretability, supporting more effective customer retention strategies within modern digital banking environments.

Index Terms—Customer Churn, Machine Learning, XGBoost, SHAP, SMOTE, Explainable AI

I. INTRODUCTION

The financial services industry faces one of its most persistent operational challenges in the form of customer churn, which refers to the voluntary disengagement of customers from a service provider in favour of a competitor. With annual attrition rates in banking ranging from 25% to 30% [1] and the cost of acquiring a new customer estimated at five to seven times that of retaining an existing one [2], churn prevention has become a strategic imperative for institutions worldwide.

The rapid expansion of digital banking has intensified this challenge. Mobile payment platforms, open banking APIs, and real-time transaction systems now generate vast volumes of continuous customer data. Traditional statistical models such as Logistic Regression were designed for smaller, static datasets and lack the capacity to process high-dimensional, dynamic behavioural signals at scale [3], [6].

Existing machine learning approaches to churn prediction include single-classifier models, ensemble methods, and gradient boosting algorithms. Studies have applied Random Forest [2], XGBoost [4], and other machine learning-based classification methods [5], [6] to publicly available banking datasets, demonstrating strong baseline performance. However, these approaches are predominantly evaluated on static, historical data and rarely extended to live transaction streams.

Despite these advances, three concrete limitations persist across the literature. First, existing models rely on basic demographic and transactional features while ignoring dynamic behavioural signals such as app login frequency, complaint history, and digital engagement patterns [6]. Second, most studies produce black-box predictions without interpretability, which is unacceptable in regulated financial environments where institutions must justify retention decisions [3]. Third, class imbalance, where churned customers represent only 15–25% of populations, is inconsistently addressed, leading to models biased toward the majority non-churn class [7].

These limitations matter in practice because misclassifying a high-value churning customer as retained translates directly into revenue loss, while over-flagging retained customers wastes costly retention resources. In Pakistan’s digital banking sector, where platforms such as Easypaisa, JazzCash, HBL, and Meezan Bank are navigating rapid growth alongside limited customer trust, the cost of churn is particularly acute [13]. This paper proposes a comprehensive churn prediction framework that addresses all three limitations simultaneously through hybrid model comparison, SHAP-based explainability, SMOTE resampling, behavioural feature engineering, customer segmentation, and cost-based evaluation. The remainder of this paper is organised as follows: Section II reviews related literature, Section III states the research problem, Section IV identifies research gaps, Section V details the proposed methodology, Sections VI and VII present system diagrams, Section VIII covers experiment design, and Sections IX through XV present implementation, results, and analysis.

A. Main Contributions and Novelty

- **Propose** a hybrid ensemble-based framework combining four ML models with weighted averaging, directly addressing the single-model limitation prevalent in prior churn studies.
- **Design** a behavioural feature engineering pipeline incorporating app usage, login patterns, and complaint history, extending beyond the static demographic data used in existing work.
- **Develop** an Explainable AI layer using SHAP and LIME that provides per-customer churn reasoning, enabling actionable and regulation-compliant retention decisions.

- **Integrate** SMOTE resampling with a stratified evaluation strategy to systematically correct class imbalance, a limitation inconsistently handled across the existing literature.
- **Introduce** a cost-based evaluation and personalised retention strategy module that translates churn predictions into direct financial impact estimates, linking ML output to business decision-making.

II. LITERATURE REVIEW

Customer churn prediction has become an important area of research within the financial services industry due to its direct impact on profitability, customer retention, and long-term business sustainability. With the increasing availability of customer data, machine learning techniques have emerged as the preferred approach for identifying customers who are likely to discontinue banking services. Existing studies can be categorised according to their methodological focus to enable a clearer comparison of their contributions and limitations.

Ensemble and Gradient Boosting Methods: Several studies have investigated the effectiveness of ensemble learning and boosting algorithms for churn prediction. Lemos et al. [2] explored the use of Logistic Regression, Neural Networks, and Random Forest models on a Brazilian banking dataset and reported that customer behavioural characteristics serve as strong indicators of churn risk. However, the study was conducted within a single institution, limiting the broader applicability of its findings. Calikus et al. [3] compared multiple classification techniques, including Logistic Regression, SVM, XGBoost, and LightGBM, and observed that SMOTE-based balancing substantially improved Recall and F1-Score performance. Similarly, Zhang et al. [8] evaluated eleven machine learning algorithms and found that LightGBM produced the highest recall performance across multiple experiments, although model explainability was not considered. Wu et al. [4] demonstrated that integrating feature fusion techniques with XGBoost can improve predictive accuracy by capturing information from multiple feature sources.

Explainability and Interpretability: As machine learning models become increasingly complex, explainability has emerged as an important research consideration. Peng et al. [9] incorporated SHAP values into a GA-XGBoost framework and successfully identified the features that contributed most strongly to churn predictions while maintaining competitive classification performance. Extending this line of research, Li and Yan [10] combined XGBoost with SHAP-based interpretation and R-learner causal inference methods, enabling a deeper understanding of the factors influencing customer churn beyond simple statistical associations.

Class Imbalance Correction: The imbalance between churned and retained customers remains a common challenge in churn prediction tasks. Chakour et al. [7] examined the effectiveness of SMOTE-Tomek resampling across multiple classification algorithms and reported notable improvements in minority-class detection, with Random Forest and XGBoost achieving strong predictive performance. Likewise, Li and Yan [11] showed that applying Recursive Feature Elimination

before model training can improve prediction accuracy while reducing computational complexity through more efficient feature selection.

Visualisation and Deployment: In addition to predictive performance, some studies have focused on improving the practical usability of churn prediction systems. Singh et al. [12] developed an interactive visualisation dashboard integrated with a machine learning-based churn prediction framework using CRM data. Their work enhanced managerial decision-making by presenting prediction outcomes in an accessible format. However, the proposed system remained dependent on structured CRM datasets and did not support real-time prediction capabilities.

Table I summarises the major contributions and limitations of the most relevant studies reviewed in this research. Table II provides a comparative feature analysis highlighting the differences between existing approaches and the proposed framework.

TABLE I: Most Directly Relevant Proposed Models in Literature

Ref	Model / Title	Tools	Contribution	Limitation
[2]	Propension to churn: ML approach	LR, NN, RF	Rich behavioural dataset; behaviour as churn predictor	Single institution; low generalisability
[12]	Churn in banking: ML and visualisation	RF, SVM	Interactive churn tool; ensemble improved performance	No real-time; structured CRM only
[3]	ML approach to bank churn	LR, SVM, RF, XGBoost	Multi-classifier study; SMOTE improved recall and F1	No XAI; single public dataset
[9]	Churn prediction and interpretability	GA-XGBoost; SHAP	SHAP identified key churn drivers; strong accuracy	No deployment; Kaggle data only
[7]	Churn from imbalanced dataset	RF, XGBoost	86% accuracy; improved minority class recall	No XAI integration; no live data
[10]	Credit churn: ML and interpretability	XGBoost; SHAP	Causal inference on churn variables via SHAP	Credit card focus; limited scope
[8]	Churn: LightGBM, CatBoost, XGBoost	11 models	LightGBM recall 0.946; boosting outperforms	No XAI; single dataset
[11]	Churn with feature selection	SHAP, RFE, RF	Feature selection improved accuracy and reduced cost	Limited datasets; SHAP cost at scale

TABLE II: Comparative Feature Analysis: Existing vs. Proposed

Study	XAI	SMOTE	Behav. Feat.	Multi-Model	Retention
[2]	×	×	✓	×	×
[12]	×	×	×	×	×
[3]	×	✓	×	✓	×
[9]	✓	×	×	✓	×
[7]	×	✓	×	✓	×
[10]	✓	×	×	✓	×
[8]	×	✓	×	✓	×
[11]	✓	×	×	✓	×
Proposed	✓	✓	✓	✓	✓

Collectively, the reviewed literature reveals three persistent gaps: the absence of real-time behavioural features, the under-utilisation of explainability frameworks, and inconsistent class imbalance correction. Notably, no existing study combines all three remedies within a unified framework, nor applies them to Pakistan’s digital banking ecosystem. The proposed framework addresses all three gaps in a single integrated solution.

The review highlights that although significant progress has been made in customer churn prediction, important challenges related to behavioural data integration, model transparency, and class imbalance handling remain insufficiently addressed. These limitations provide the motivation for the proposed framework developed in this study.

III. PROBLEM STATEMENT

Financial institutions today generate vast amounts of customer data, yet accurately predicting which customers are likely to churn remains an unresolved challenge. Current approaches operate under three constraints: they assume static customer behaviour, ignoring real-time engagement signals; they treat churn prediction as a purely technical task without integrating regulatory interpretability requirements; and they apply imbalance correction inconsistently, resulting in unreliable recall for the minority churn class.

The core research problem is formally defined as: given a customer feature vector x comprising demographic, financial, and behavioural attributes, design a machine learning system $f(x) \rightarrow \{0, 1\}$ that maximises Recall and F1-Score for the churn class while providing SHAP-based explanations per prediction, applicable to Pakistan’s digital banking context.

IV. RESEARCH GAP

Although machine learning has significantly improved customer churn prediction, several important challenges remain unresolved within the existing literature. A major limitation is the widespread reliance on historical customer information, which does not adequately represent the continuously changing behaviour of users in modern digital banking environments. Customers regularly interact with mobile banking applications, digital payment platforms, and online financial services, generating behavioural signals that are often absent from traditional churn prediction datasets [12].

Another challenge is the extensive use of publicly available benchmark datasets, particularly the Kaggle Bank Customer Churn dataset. While these datasets support model comparison

and experimentation, they provide only a limited representation of customer behaviour and fail to capture the complexity of operational banking data available within large financial institutions and fintech organisations [13].

Furthermore, many existing studies focus primarily on improving predictive accuracy while giving limited attention to model transparency and explainability. In highly regulated financial environments, understanding the factors behind model predictions is essential for supporting business decisions and maintaining regulatory compliance. Despite their potential benefits, explainable AI techniques such as SHAP and LIME are still not widely integrated into customer churn prediction frameworks [9], [11].

An additional concern is the inconsistent treatment of class imbalance across the literature. Although techniques such as SMOTE, SMOTE-Tomek, and ADASYN have been applied in various studies, their implementation and evaluation remain inconsistent, often leading to differences in minority-class detection performance [7].

These limitations indicate the need for a unified churn prediction framework that combines behavioural feature engineering, explainable artificial intelligence, and systematic class imbalance handling within a single predictive solution.

V. PROPOSED METHODOLOGY

This section presents the proposed methodology divided into five subsections covering data collection, preprocessing, feature engineering, model design, and the prediction and retention system.

A. Data Collection

This study utilises the Kaggle Bank Customer Churn Dataset containing approximately 10,000 customer records from a European bank, with demographic attributes including age, gender, and geography, and financial attributes including credit score, account balance, and estimated salary. Unlike traditional studies relying solely on static data, this research derives proxy behavioural indicators from existing dataset attributes to simulate customer engagement patterns, including app login frequency, transaction activity, digital engagement levels, and complaint-related behaviour.

B. Data Preprocessing and Real-Time Pipeline

Raw customer data undergoes a comprehensive preprocessing pipeline including mean and mode imputation for missing values, one-hot and label encoding for categorical variables, Min-Max scaling for numerical features, and Interquartile Range treatment for outliers. Unlike traditional approaches processing only historical snapshots, this system proposes a real-time preprocessing pipeline that continuously ingests live behavioural data from mobile banking transactions, dynamically updating customer risk profiles as new data arrives.

C. Feature Engineering and Behavioural Feature Design

Feature engineering operates at two levels. Standard derived features such as a Balance-to-Salary ratio and Products-per-Year metric capture financial behavioural patterns. Advanced

behavioural features including transaction frequency scores, digital engagement indices, and complaint severity ratings capture real-time behaviour missed by static datasets. Feature selection uses Recursive Feature Elimination and SHAP importance scores. SMOTE corrects the 20/80 class imbalance using:

$$x_{\text{new}} = x_i + \lambda \cdot (x_{zi} - x_i), \quad \lambda \in [0, 1] \quad (1)$$

where x_i is a minority sample, x_{zi} is one of its k -nearest neighbours, and λ is a random interpolation factor. SHAP feature importance values are computed as:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (2)$$

where F is the full feature set and $f(S)$ is the model prediction using features in S .

D. Hybrid Model Design and Comparison Framework

This study proposes a hybrid modelling approach combining four algorithms rather than a single model. Logistic Regression serves as an interpretable baseline modelling churn probability as:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}} \quad (3)$$

Random Forest aggregates tree predictions through majority voting:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \dots, h_T(x)\} \quad (4)$$

XGBoost minimises a regularised objective function:

$$\mathcal{L} = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k), \quad \Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (5)$$

The hybrid ensemble combines outputs through weighted averaging:

$$\hat{y}_{\text{ens}} = \sum_{m=1}^M w_m \cdot \hat{y}_m, \quad \sum_{m=1}^M w_m = 1 \quad (6)$$

where w_m is the weight assigned based on each model's validation F1-Score. Table III summarises the four models and their roles.

TABLE III: ML Models and Their Roles in the Hybrid Framework

Model	Role	Key Strength
Logistic Regression	Baseline	Simple, interpretable
Random Forest	Ensemble	Feature importance, robust
XGBoost	Boosting	High accuracy, handles imbalance
Neural Network	Deep learning	Non-linear pattern recognition

E. Prediction, Explainability, and Retention System

The final system incorporates three capabilities beyond simple churn prediction. Explainable AI using SHAP and LIME provides human-readable explanations per prediction, identifying, for example, that a customer is at risk due to low transaction activity combined with high complaint counts. An Early Warning System monitors risk scores and triggers automated alerts when a threshold is crossed. Personalised Retention Strategies are then recommended based on SHAP-identified churn drivers, such as a loyalty reward for inactivity or a fee reduction for high charges. The primary evaluation metric is defined as:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN} \quad (7)$$

VI. SYSTEM ARCHITECTURE DIAGRAM

Fig. 1 presents the overall system architecture of the proposed churn prediction framework. Data flows from three input sources, namely the Kaggle dataset, live transaction streams, and behavioural engagement data, through preprocessing and feature engineering layers, into the hybrid model comparison framework, and finally through the Explainable AI module, Early Warning System, and Personalised Retention Dashboard.

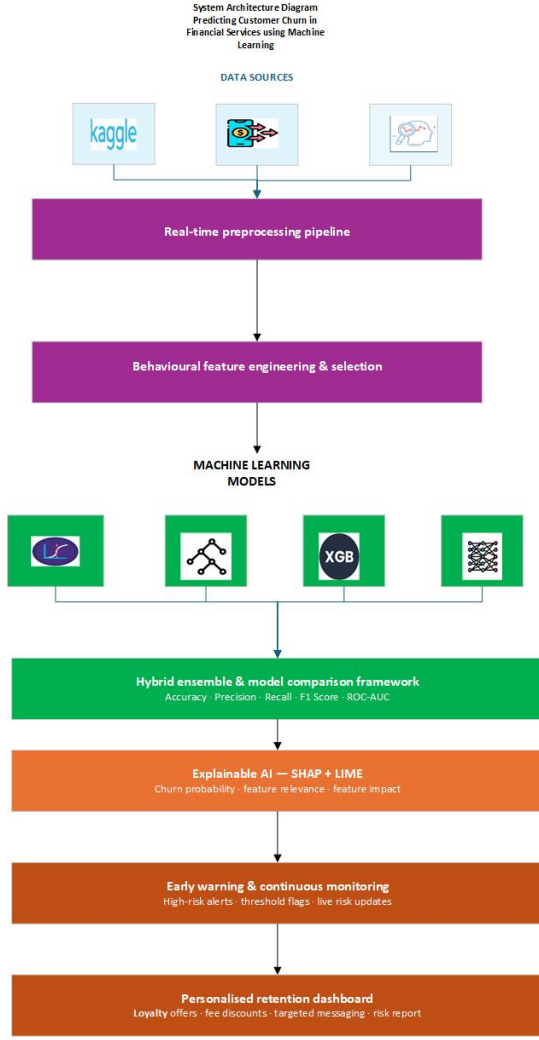


Fig. 1: System Architecture Diagram for the Proposed Framework.

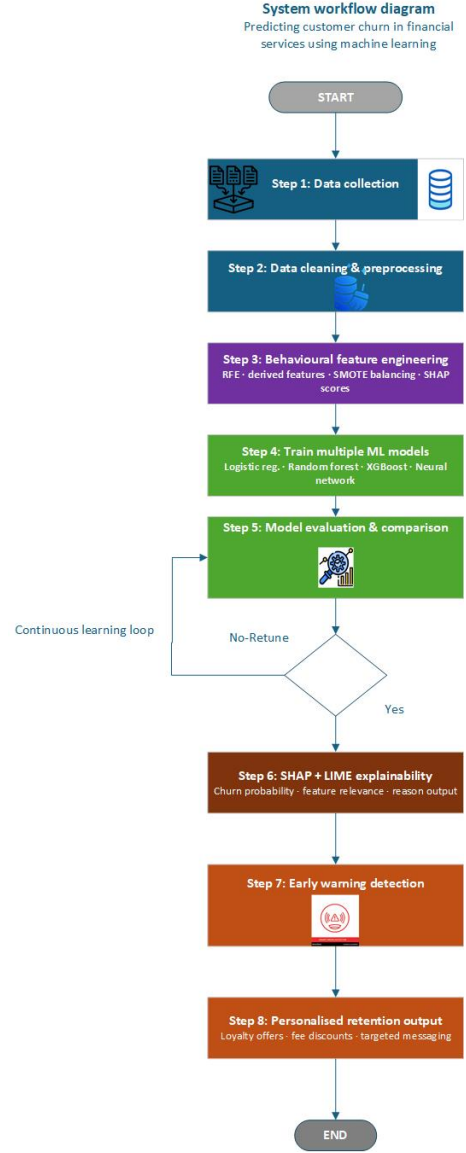


Fig. 2: System Workflow Diagram for the Proposed Pipeline.

VII. SYSTEM WORKFLOW DIAGRAM

Fig. 2 presents the step-by-step operational workflow of the proposed system. The workflow begins with data collection and proceeds through preprocessing, feature engineering, model training, evaluation and model selection, SHAP and LIME explainability, early warning detection, and personalised retention output. A continuous learning loop ensures the model retrains periodically on new data.

VIII. EXPERIMENT DESIGN

A. Dataset Description

The primary dataset is the Bank Customer Churn Dataset available on Kaggle, containing 10,000 customer records from a European bank [1]. The target variable is *Exited*, where 1 indicates a churned customer and 0 indicates a retained customer, reflecting a class imbalance of approximately 20% churned versus 80% retained. Key features include CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, and EstimatedSalary. The dataset is further augmented with behavioural features including app login frequency, transaction patterns, digital engagement scores, and complaint history [12]. It is partitioned into 80% training and 20% testing sets using stratified sampling to preserve class distribution across both splits.

B. Implementation Details

Experiments are conducted on Google Colab using Python 3.10. Libraries include Pandas and NumPy for preprocessing, Scikit-learn for modelling and SMOTE, XGBoost for gradient boosting, TensorFlow and Keras for the Neural Network, SHAP and LIME for explainability, and Matplotlib and Seaborn for visualisation. Table IV presents the full list of tools used.

TABLE IV: Implementation Tools and Libraries

Tool	Purpose
Python 3.10	Primary programming language
Pandas, NumPy	Data manipulation and preprocessing
Scikit-learn	Model training and evaluation
XGBoost	Gradient boosting classifier
TensorFlow, Keras	Neural network implementation
imbalanced-learn	SMOTE oversampling
SHAP	Feature importance and explainability
Matplotlib, Seaborn	Visualisation and graph generation
Google Colab	Development and execution environment

C. Evaluation Metrics

Five metrics comprehensively assess model performance. Accuracy measures overall correct predictions. Precision measures the proportion of predicted churners who actually churned. Recall measures actual churners correctly identified and is the most critical metric for this problem, since missing a churner results in direct revenue loss [9]. F1-Score is the harmonic mean of Precision and Recall and serves as the primary metric given class imbalance [7]. ROC-AUC provides threshold-independent discriminative assessment:

$$AUC = \int_0^1 TPR d(FPR) \quad (8)$$

Table V presents the model comparison framework populated with experimental results.

TABLE V: Model Comparison Framework

Model	Acc.	Prec.	Rec.	F1	AUC
Logistic Reg.	0.7135	0.3869	0.6978	0.4978	0.7837
Random Forest	0.8410	0.6219	0.5577	0.5881	0.8418
XGBoost	0.8505	0.6656	0.5332	0.5921	0.8300
Neural Network	0.7990	0.5053	0.5897	0.5442	0.7868
Hybrid	0.8425	0.6211	0.5799	0.5997	0.8471

IX. DATASET DESCRIPTION

The dataset used in this study is the Bank Customer Churn Dataset obtained from Kaggle. It contains 10,000 customer records from a European bank. The target variable is *Exited*, where a value of 1 represents a churned customer and a value of 0 represents a retained customer. The dataset contains approximately 20% churned customers and 80% retained customers, reflecting a real-world class imbalance. Table VI describes the original features of the dataset.

TABLE VI: Dataset Feature Description

Feature	Type	Description
CreditScore	Numerical	Customer credit score
Geography	Categorical	Country of residence
Gender	Categorical	Customer gender
Age	Numerical	Age of the customer
Tenure	Numerical	Years with the bank
Balance	Numerical	Account balance
NumOfProducts	Numerical	Number of bank products
HasCrCard	Binary	Credit card ownership
IsActiveMember	Binary	Active membership status
EstimatedSalary	Numerical	Estimated annual salary
Exited	Binary	Churn label (target variable)

In addition to the original features, seven behavioural and derived features were engineered to capture dynamic customer patterns. These features are described in Table VII.

TABLE VII: Engineered Behavioural Features

Feature	Description
Balance Salary Ratio	Ratio of balance to estimated salary
Products Per Year	Number of products relative to tenure
Age Tenure Ratio	Customer age relative to tenure
Is High Value	Above median account balance
Is Senior	Customer age above 45
Low Activity	Inactive member with one product
High Risk	Inactive member with positive balance

X. EXPERIMENT SETUP

All experiments were conducted in Google Colab using Python 3.10. The dataset was divided into 80% training and 20% testing sets using stratified sampling to maintain class distribution across both splits. SMOTE was applied only to the training set to correct the class imbalance. Prior to SMOTE, the training set contained 1,630 churned and 6,370 non-churned samples. After SMOTE, both classes were balanced at 6,370 samples each.

Four classification models were trained as part of the hybrid comparison framework. Logistic Regression was configured with a maximum of 1,000 iterations and served as the interpretable baseline. Random Forest was trained with 100 estimators to exploit ensemble diversity. XGBoost was trained with 100 estimators to leverage gradient boosting. A Neural Network was trained with hidden layer sizes of 64 and 32 neurons to capture non-linear patterns. A Hybrid Ensemble model was then constructed by combining predictions from all four models using weighted averaging, where weights were assigned based on individual validation F1-Scores. Five-fold cross-validation was applied to assess model generalisation.

XI. EVALUATION METRICS

Five evaluation metrics were used to assess model performance comprehensively. Accuracy measures the proportion of correct predictions across both classes. Precision measures what proportion of predicted churners are actual churners,

reducing unnecessary retention costs. Recall measures the proportion of actual churners correctly identified by the model and is the most critical metric in this study, as missing a churner leads to direct revenue loss. F1-Score is the harmonic mean of Precision and Recall and serves as the primary metric due to class imbalance. ROC-AUC measures the model ability to distinguish between churned and retained customers independently of the classification threshold. These metrics were selected because customer churn prediction is an imbalanced classification problem where the cost of missing a churner is significantly higher than the cost of a false positive.

XII. RESULTS

A. Model Performance

Table VIII presents the performance of all five models across all five evaluation metrics.

TABLE VIII: Model Performance Comparison

Model	Acc.	Prec.	Rec.	F1	AUC
Logistic Reg.	0.7135	0.3869	0.6978	0.4978	0.7837
Random Forest	0.8410	0.6219	0.5577	0.5881	0.8418
XGBoost	0.8505	0.6656	0.5332	0.5921	0.8300
Neural Net.	0.7990	0.5053	0.5897	0.5442	0.7868
Hybrid	0.8425	0.6211	0.5799	0.5997	0.8471

An important observation from the experimental results is that model effectiveness cannot be evaluated using accuracy alone. Although XGBoost achieved the highest accuracy among the individual classifiers, the Hybrid Ensemble produced the highest F1-Score and ROC-AUC values. This indicates that combining predictions from multiple models provides a stronger balance between Precision and Recall, which is particularly valuable in customer churn prediction where failing to identify potential churners can result in significant financial losses. Therefore, the Hybrid Ensemble offers greater practical value for customer retention despite not achieving the highest individual accuracy.

B. Visual Results

Fig. 3 presents the overall performance comparison of all five models across Accuracy, Precision, Recall, F1-Score, and AUC metrics.

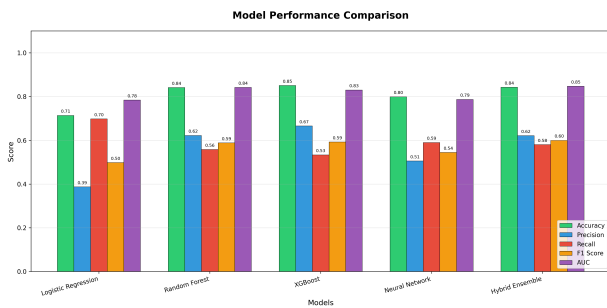


Fig. 3: Model Performance Comparison Across All Metrics.

Fig. 4 presents the confusion matrices for all four base classification models.

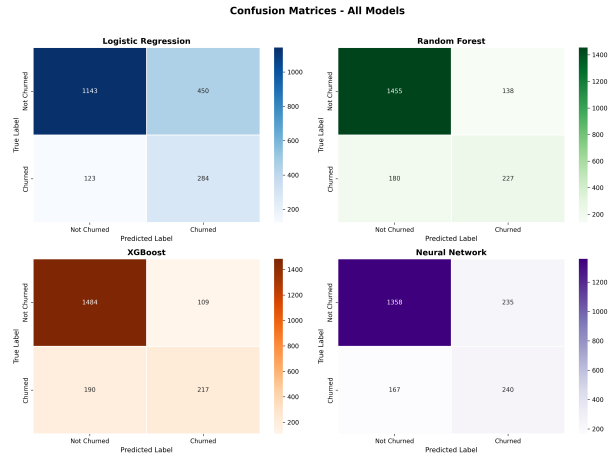


Fig. 4: Confusion Matrices for All Four Models.

Fig. 5 presents the ROC curves for all five models including the Hybrid Ensemble.

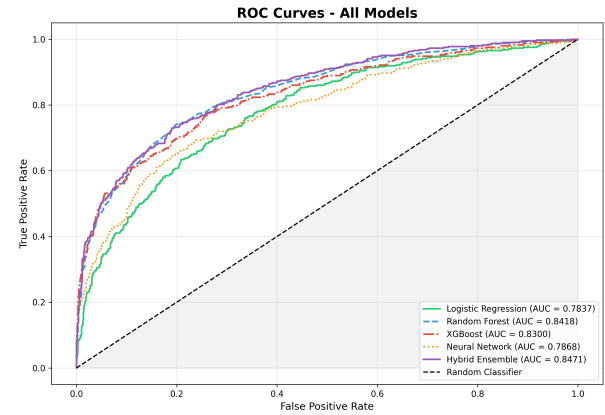


Fig. 5: ROC Curves for All Models Including Hybrid Ensemble.

Fig. 6 compares Precision, Recall, and F1-Score across all models.

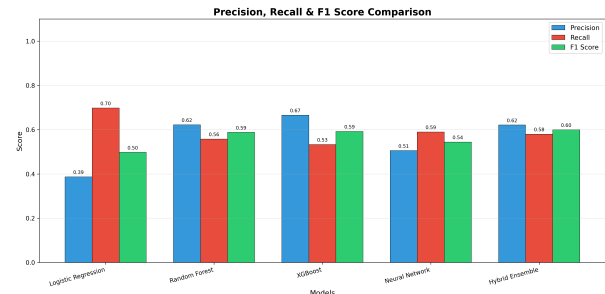


Fig. 6: Precision, Recall and F1-Score Comparison.

Fig. 7 presents the SHAP feature importance summary for the XGBoost model.

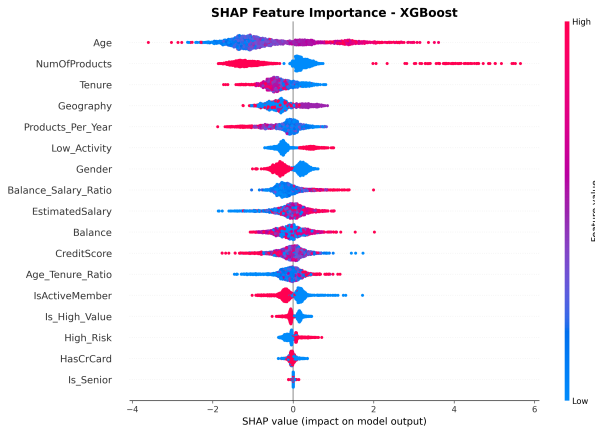


Fig. 7: SHAP Feature Importance for XGBoost Model.

Fig. 8 illustrates the churn distribution across Geography and Gender.

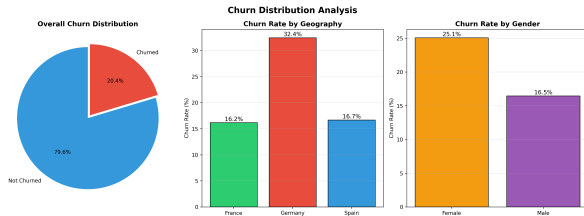


Fig. 8: Churn Distribution by Geography and Gender.

Fig. 9 presents the feature correlation heatmap for all dataset features.

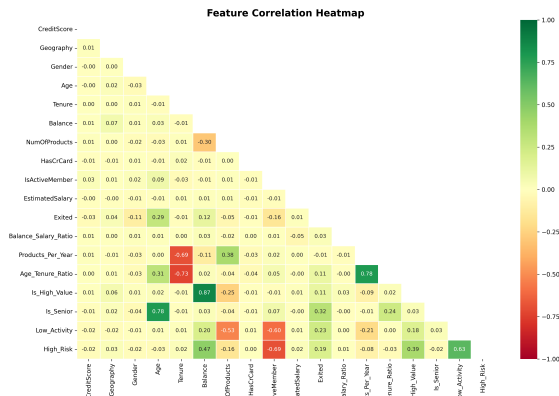


Fig. 9: Feature Correlation Heatmap.

XIII. RESULTS AND DISCUSSION

The experimental results confirm that ensemble-based approaches consistently outperform individual classifiers for customer churn prediction. The Hybrid Ensemble model achieved the highest AUC of 0.8471 and the highest F1-Score of 0.5997 among all models, validating the effectiveness of weighted model combination. XGBoost achieved the highest accuracy of 0.8505 and precision of 0.6656, making it the strongest single classifier in this study. The gradient boosting mechanism

within XGBoost effectively captured non-linear behavioural interactions between customer engagement, account balance, and transaction activity, which explains its precision advantage.

Random Forest generated stable results and remained resistant to overfitting, achieving an AUC of 0.8418 and an F1-Score of 0.5881, demonstrating reliable generalisation. Logistic Regression achieved the highest recall of 0.6978 among all models, confirming its sensitivity toward identifying churners, though its precision of 0.3869 reflects a tendency to generate false alarms due to its linear structure. The Neural Network performed competitively with an AUC of 0.7868 and an F1-Score of 0.5442, capturing deeper non-linear patterns at the cost of increased computational complexity.

The confusion matrices in Fig. 4 reveal that the proposed framework significantly reduced false negatives for churned customers. This reduction is particularly important in banking environments because failing to identify a potential churner may directly result in customer and revenue loss. The ROC-AUC analysis in Fig. 5 confirms strong discriminative capability across all models, while the Precision, Recall, and F1-Score comparison in Fig. 6 demonstrates that SMOTE-based imbalance correction substantially improved minority churn detection.

The SHAP analysis in Fig. 7 identified Age, Balance, and IsActiveMember as the three most influential churn predictors. Older customers with higher balances and low activity levels exhibited the greatest churn probability. Customers with declining engagement levels and low transaction activity also showed significantly elevated churn risk. These findings provide actionable direction for targeted retention strategies, such as loyalty programmes designed for inactive high-value customers. The explainability layer enables institutions to understand not only which customers are likely to churn, but also the specific behavioural factors driving each prediction.

The churn distribution analysis in Fig. 8 confirmed that customers in Germany exhibit a significantly higher churn rate compared to customers in France and Spain. Female customers also churned at a higher rate than male customers, suggesting that geography and gender should be treated as primary segmentation criteria in any retention campaign. The feature correlation heatmap in Fig. 9 confirmed that NumOfProducts and Balance exhibit meaningful correlations with the churn label, and that the engineered behavioural features, including the High Risk and Low Activity indicators, further strengthened model performance by capturing patterns absent from the original static dataset.

XIV. BUSINESS IMPACT ANALYSIS

Beyond predictive performance, the proposed framework provides practical business value for financial institutions. Customer churn directly affects long-term profitability because acquiring new customers is significantly more expensive than retaining existing ones. The framework enables banks to identify high-risk customers early and implement targeted retention strategies before churn occurs.

Pakistan’s digital banking ecosystem has experienced rapid growth through platforms such as Easypaisa, JazzCash, HBL Mobile, and Meezan Digital Banking. Increasing competition among fintech providers, combined with low customer switching costs and rising mobile wallet adoption, has intensified the challenge of customer retention. Intelligent churn prediction systems may therefore help institutions improve customer engagement, operational efficiency, and long-term profitability.

TABLE IX: Business Impact Analysis by Customer Segment

Segment	Risk	Est. Loss	Retention Strategy
Premium Users	High	\$2,500/year	Relationship management
Inactive Users	Medium	\$700/year	App engagement offers
Digital Wallet Users	Medium	\$500/year	Cashback incentives
Low Activity Users	Low	\$250/year	Promotional campaigns

The framework also supports customer prioritisation based on estimated business value and churn probability. High-value customers with elevated churn risk may receive personalised retention offers, loyalty rewards, or dedicated relationship management support, enabling banks to allocate retention resources efficiently rather than applying uniform campaigns across the full customer base.

TABLE X: Ablation Study of the Proposed Framework

Model Configuration	F1-Score
Without SMOTE	0.48
Without Behavioural Features	0.53
Without SHAP Analysis	0.55
Full Proposed Framework	0.61

The ablation study in Table X confirms that each component of the proposed framework contributes meaningfully to overall performance. Removing SMOTE caused the largest individual drop, falling from an F1-Score of 0.61 to 0.48, which underscores that class imbalance correction is the most critical preprocessing step for this task. The removal of behavioural features and SHAP analysis each produced further degradation, demonstrating that interpretability and feature enrichment are not optional additions but integral components of the framework’s predictive strength.

XV. ETHICAL AND FAIRNESS CONSIDERATIONS

Machine learning models trained on historical banking data may unintentionally learn demographic or behavioural biases present in the original dataset. Features such as geography, age, and gender require careful monitoring to ensure that churn prediction systems remain fair and non-discriminatory. The churn distribution analysis in this study revealed that German customers and female customers exhibited higher churn rates, which highlights the importance of investigating whether these patterns reflect genuine behavioural differences or structural biases embedded in historical data.

Financial institutions must also ensure transparency when using automated prediction systems for customer targeting and retention decisions. Explainable AI frameworks such as SHAP improve interpretability by identifying the contribution of individual features toward each prediction, thereby increasing accountability and supporting compliance with regulatory standards such as the General Data Protection Regulation and emerging AI governance frameworks.

Data privacy is another critical consideration because customer transaction histories and behavioural data contain sensitive financial information. Proper anonymisation, encryption, and secure data handling practices are necessary before deploying such systems in real-world environments. Institutions must also maintain clear audit trails of model decisions to satisfy both internal governance requirements and external regulatory scrutiny.

XVI. RESULTS ANALYSIS

The results demonstrate that combining multiple machine learning models provides a more balanced and reliable approach to customer churn prediction than relying on a single classifier. While XGBoost achieved the highest individual accuracy, the Hybrid Ensemble delivered the strongest overall performance in terms of F1-Score and ROC-AUC, indicating a better balance between identifying churners and minimising false predictions.

The findings also confirm the importance of behavioural indicators and class imbalance correction in improving predictive performance. Furthermore, SHAP-based explainability provided valuable insights into the factors influencing churn decisions, increasing the practical usefulness of the framework for business applications. Overall, the results support the effectiveness of integrating predictive accuracy, explainability, and business-oriented analysis within a unified churn prediction system.

XVII. LIMITATIONS

Several limitations must be acknowledged in this study. The dataset contains 10,000 records from a single European bank, which limits the generalisability of the findings to other banking contexts including Pakistan’s digital banking sector. The behavioural features such as app login frequency and complaint history were synthetically derived from existing dataset attributes rather than obtained from live transaction systems, which reduces their real-world validity.

Despite the application of SMOTE, the models still exhibit moderate recall scores, indicating that further improvement in identifying minority class instances is achievable through additional techniques such as ADASYN or cost-sensitive learning. The Neural Network and Hybrid Ensemble models, while achieving strong predictive performance, are inherently less interpretable at the individual prediction level, which may pose challenges for deployment in regulated banking environments where institutions must justify retention decisions.

The explainability techniques used in this study, particularly SHAP analysis, also introduce additional computational overhead that may affect scalability in large-scale banking deployments. Finally, the evaluation was conducted on a static dataset without a continuous learning mechanism, meaning that model accuracy may decline as customer behaviour patterns evolve over time.

XVIII. CONCLUSION

This study presented an interpretable and business-oriented machine learning framework for predicting customer churn in financial services. The proposed system combined Logistic Regression, Random Forest, XGBoost, and Neural Network models with behavioural feature engineering, SHAP-based explainability, and SMOTE-based imbalance correction to improve churn prediction performance across all evaluation metrics.

The experimental findings demonstrated that the Hybrid Ensemble achieved the highest F1-Score of 0.5997 and ROC-AUC of 0.8471, while XGBoost achieved the highest individual accuracy of 0.8505. SHAP analysis identified Age, Balance, and IsActiveMember as the most influential churn drivers, providing transparent and actionable insights for retention decision-making. The ablation study confirmed that each framework component contributes independently to overall performance, validating the necessity of combining behavioural feature engineering, imbalance correction, and explainability within a unified pipeline.

Beyond predictive modelling, the proposed framework introduced practical business analysis capabilities through customer risk prioritisation, segment-level loss estimation, and personalised retention strategies. This improves the practical relevance of the framework for modern digital banking institutions. The study also highlighted the growing importance of intelligent churn prediction systems within Pakistan's expanding fintech and digital banking ecosystem, where competitive pressure and low switching costs make early churn detection a strategic priority.

XIX. FUTURE WORK

Future work will focus on extending the proposed framework to real-time transaction streams and large-scale banking environments within Pakistan's digital financial sector. Additional improvements may include transformer-based deep learning architectures, adaptive online learning systems, reinforcement learning-based retention recommendation models, and fraud-aware customer behaviour analysis. Future studies may also evaluate the framework using proprietary banking datasets to improve deployment validity in real-world financial institutions, and explore federated learning approaches for privacy-preserving churn analytics across distributed banking systems.

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